

SPL Exercise Aggregate functions and grouping

1. COUNT() function

```
SELECT COUNT(*) AS total_employees  
FROM employees;
```

Results

total_employees
10

2. SUM() function

```
SELECT SUM(salary) AS total_it_salary  
FROM employees  
WHERE department = 'IT';
```

Results

$53\ 000 + 53\ 000 + 62\ 000 + 50\ 000$
 $= 220\ 000$

total_it_salary
220 000

3. AVG function

```
SELECT AVG(salary) AS avg_hr_salary  
FROM employees  
WHERE dept department = 'IT';
```


Calculations

$$\begin{aligned} &48000 + 51000 \\ &= 99000 \\ &\quad 2 \\ &= 49500 \end{aligned}$$

Results

avg - hr salary
49500

4. Min and Max functions

SELECT MIN(salary) AS lowest-salary,
MAX(salary) AS highest-salary
FROM employees;

Results

lowest salary	highest salary
48000	62000

5. Group By Statement

SELECT department, SUM(salary) AS total-salary
FROM employees
GROUP BY department

Results

department	total - salary
IT	22 000
HR	11 000
Finance	119 000
Marketing	105 000

6. GROUP BY and COUNT ()

SELECT city, COUNT (*) AS employee_count
 FROM employees
 GROUP BY city ;

city	employee_count
New York	2
Chicago	4
Los Angeles	2
San Francisco	2
Houston	1

7. GROUP BY and ORDER BY

SELECT department,
 AVG (salary) AS avg_salary
 FROM employees
 GROUP BY department
 ORDER BY avg_salary DESC ;

Results

Department	Sum Salary
Finance	59500
IT	5600
Marketing	53500
HR	49500

5. HAVING clause

```
SELECT department,  
SUM(Salary) AS total_salary  
FROM employees  
GROUP BY department  
HAVING SUM(Salary) > 100000
```

Results

Department	total salary
IT	22000
Finance	118000
Marketing	105000

9. GROUP BY HAVING and ORDER BY

```
SELECT city,  
(COUNT(*) AS employee_count)  
FROM employees  
GROUP BY city  
HAVING (COUNT(*) > 1)  
ORDER BY employee_count DESC
```


Results

City	employee count
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10. Department with the highest average salary

```

SELECT department
      Avg. (Salary) AvgSalary
FROM employees
GROUP BY department
ORDER BY avgSalary Desc
Limit 1;
  
```

department	avgSalary
finance	59500